

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

Q.1: B — Electric current 1

Explanation: The SI system has seven **base (fundamental) quantities**: length, mass, time, electric current, temperature, amount of substance, and light intensity. Force, density, and velocity are all **derived quantities**, obtained by combining base quantities (e.g. force = mass $\times$ acceleration).

Q.2: C — $\text{kg}/(\text{m}\cdot\text{s}^2)$ 1

Explanation: Pressure = force / area = $(\text{kg}\cdot\text{m}/\text{s}^2) / \text{m}^2 = \text{kg}/(\text{m}\cdot\text{s}^2)$. Option B ($\text{kg}\cdot\text{m}/\text{s}^2$) is the dimension of force (newton) alone, not pressure; students often forget to divide by the area term.

Q.3: C — Instantaneous speed 1

Explanation: The **gradient of a distance-time graph** gives the speed of the body at that instant. Acceleration is instead given by the gradient of a speed-time graph, and displacement is a separate vector quantity read directly off the vertical axis of a displacement-time graph, not from the slope.

Q.4: D — 70 m/s 1

Explanation: Using $v = u + gt$ with $u = 0$: $v = 0 + (10)(7) = 70 \text{ m/s}$. Option A (17 m/s) wrongly adds g and t instead of multiplying; option B (35 m/s) mistakenly uses half of gt ; option C (700 m/s) forgets that g is 10, not 100.

Q.5: D — Strong force 1

Explanation: The **strong nuclear force** is the strongest of the four fundamental forces, but has an extremely short range (about the diameter of a proton). It holds protons together in the nucleus, overcoming their electromagnetic repulsion. Gravity is by far the **weakest** of the four.

Q.6: C — Newton's 3rd law 1

Explanation: Newton's third law states that **whenever one object exerts a force on a second object, the second object exerts an equal and opposite force on the first**. The feet pushing down on the ground (action) and the ground pushing up on the feet (reaction) is the textbook's own example of this law.

Q.7: A — 17.5 N·m 1

Explanation: Torque = force $\times$ perpendicular distance = $35 \text{ N} \times 0.5 \text{ m} = 17.5 \text{ N}\cdot\text{m}$. Option B doubles the value instead of halving; option C mistakenly divides distance by force instead of multiplying.

Q.8: B — Unstable 1

Explanation: A body is in **unstable equilibrium** if it moves further away from its previous position when slightly displaced and released, exactly as a pencil balanced on its tip topples over rather than returning upright. A stable body (like a book tilted slightly) returns to its original position, and a neutral body (like a ball rolled on a flat surface) simply stays in its new position.

Q.9: B — Isobars 1

Explanation: On weather maps, lines joining places with the **same atmospheric pressure** are called **isobars**. Isotherms join points of equal temperature, contours join points of equal height, and isotopes are a chemistry term for atoms of the same element with different neutron numbers.

Q.10: C — 70,000 Pa 1

Explanation: $P = \rho gh = 1000 \times 10 \times 7 = \mathbf{70,000 \text{ Pa}}$. Option A forgets to multiply by g ; option B forgets both ρ and the correct power of ten; option D mistakenly adds an extra zero.

Q.11: C — 150 J 1

Explanation: Work done at an angle is $W = Fd \cos\theta = 60 \times 5 \times 0.5 = \mathbf{150 \text{ J}}$. Option A (300 J) is the common mistake of using $W = Fd$ directly and forgetting the $\cos\theta$ factor that accounts for the force not being fully in the direction of motion.

Q.12: B — Cosmology 1

Explanation: **Cosmology** is defined in the chapter as the study of the origin, development, structure, history and future of the entire universe, from birth to death. Astronomy studies celestial objects, astrophysics applies physics laws to interpret astronomical observations, and thermal physics studies heat and the kinetic theory of gases.

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)**2(i) SI Base Quantities and Derived Units** 3

The seven **SI base quantities** are: length (metre), mass (kilogram), time (second), electric current (ampere), temperature (kelvin), amount of substance (mole), and light intensity (candela). 1

(a) **Area:** unit $\text{m} \times \text{m} = \text{m}^2$ (from base unit of length). 1

(b) **Velocity:** unit m/s (length divided by time). (c) **Force:** unit $\text{kg}\cdot\text{m/s}^2$, given the special name **newton (N)**. 1

OR**Physical vs Non-Physical Quantities** 3

Physical quantities are quantities which can be measured, whereas **non-physical quantities** are those which cannot be measured. 1

Examples of physical quantities: **length, mass, time, density, temperature** (any two). 1

Examples of non-physical quantities: **taste, feeling, colour, beauty, honesty** (any two). 1

2(ii) Speed vs Velocity 3

Speed is the distance covered per unit time; it is a **scalar** quantity (magnitude only). **Velocity** is the displacement covered per unit time; it is a **vector** quantity (magnitude and direction). 1

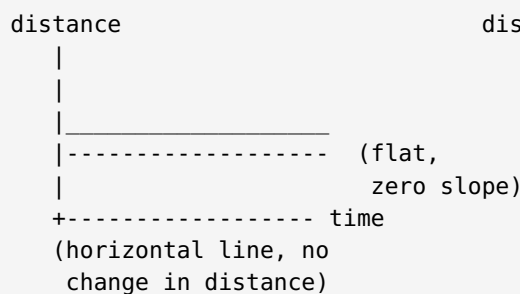
Even though the car's **speed stays constant** at 40 m/s, its **direction is continuously changing** as it goes around the circular track. 1

Since velocity is a vector that depends on direction as well as magnitude, a changing direction means the **velocity is NOT constant**, even though the speed (magnitude only) is constant. 1

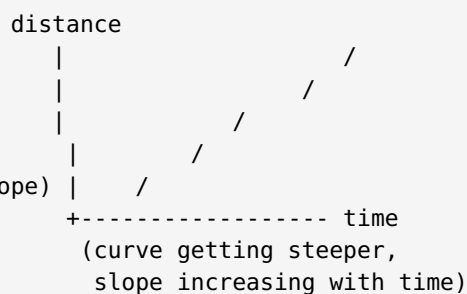
OR

Distance-Time Graphs 3

(a) Body at rest:



(b) Uniformly increasing speed:



1 for each correctly shaped graph, 1 for the following statement:

The **gradient of a distance-time graph** represents the **speed** of the object at that point.

2(iii) Equations of Motion (Van) 3

Given: $u = 8 \text{ m/s}$, $v = 32 \text{ m/s}$, $t = 6 \text{ s}$

(a) $a = (v - u)/t = (32 - 8)/6 = 24/6 = 4 \text{ m/s}^2$ 1.5

(b) $s = ut + \frac{1}{2}at^2 = (8)(6) + \frac{1}{2}(4)(6)^2 = 48 + 72 = 120 \text{ m}$ 1.5

OR

Free Fall (Drone Package) 3

Given: $h = 320 \text{ m}$, $u = 0$, $g = 10 \text{ m/s}^2$

(a) $h = \frac{1}{2}gt^2 \Rightarrow 320 = \frac{1}{2}(10)t^2 \Rightarrow t^2 = 64 \Rightarrow t = 8 \text{ s}$ 1.5

(b) $v = gt = (10)(8) = 80 \text{ m/s}$ 1.5

2(iv) Fundamental Forces 3

The four fundamental forces of nature, in **decreasing order of relative strength**, are: 2

1. **Strong nuclear force** (strongest, but shortest range)
2. **Electromagnetic force**
3. **Weak nuclear force**
4. **Gravitational force** (weakest of all four)

The **strong nuclear force** keeps protons tightly bound inside the nucleus of an atom by overcoming the electromagnetic repulsion between the positively charged protons. 1

OR

Newton's Second Law — Mass Calculation 3

Newton's second law: the acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass: $F = ma$. 1

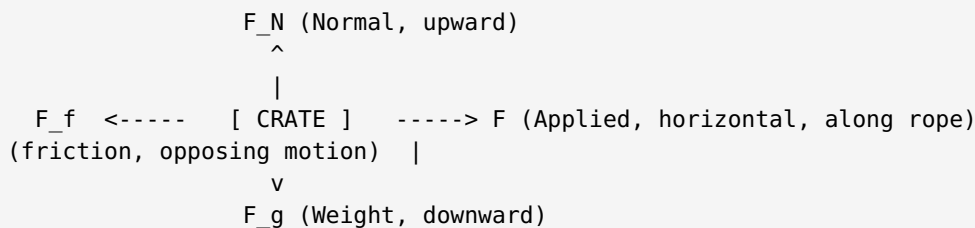
Given: $F = 45 \text{ N}$, $a = 5 \text{ m/s}^2$

$m = F/a = 45/5 = 9 \text{ kg}$ 2

2(v) System Diagram vs Free-Body Diagram 3

A **system diagram (SD)** is a visual expression showing all the objects involved in a situation. A **free-body diagram (FBD)** shows only the single object being analysed, with arrows representing all the forces acting on it. 1

FBD of crate pulled at constant velocity across a rough floor:



F_N balances F_g (vertical equilibrium);

F applied balances F_f (constant velocity, horizontal equilibrium)

2 for a correctly labelled FBD showing weight, normal force, applied force and friction, all acting on the crate alone.

OR

Net Force — Like Parallel Forces 3

Net (resultant) force: the single force that has the same effect as the combined effect of all the forces acting on a body; found by vector addition of the individual forces. 1

Since both people push the car in the **same direction**, these are **like parallel forces**, and their magnitudes simply add: 1

Net force = 18 N + 25 N = **43 N** (in the direction of push) 1

2(vi) Types of Equilibrium 3

(a) **Ball in a bowl:** when displaced slightly and released, it rolls back to the bottom of the bowl — this is **stable equilibrium**, because its weight creates a restoring moment that returns it to its original position. 1

(b) **Pencil on its tip:** when displaced slightly, it topples over completely instead of returning — this is **unstable equilibrium**, because the centre of mass moves further away from its original position once displaced. 1

(c) **Ball rolled on a flat surface:** after displacement it neither returns nor moves further, but simply stays in its new position — this is **neutral equilibrium**, because there is no moment to increase or decrease the displacement. 1

OR

Terminal Velocity 3

Terminal velocity is the maximum, constant velocity reached by a falling object when the upward drag (air resistance) force becomes equal in magnitude to its downward weight, so the net force (and acceleration) becomes zero. 1

Initially, a skydiver's **weight is much greater** than air resistance, so she accelerates downward and her speed increases rapidly. 1

As her speed increases, the **drag force also increases** (it depends on speed), until it exactly balances her weight; at this point she is in dynamic translational equilibrium and moves at a constant **terminal velocity** with zero acceleration. 1

2(vii) Friction: Advantages and Disadvantages 3

Advantages (any two): friction between the soles of shoes/feet and the ground helps us walk; friction between tyres and road helps drive/stop cars; friction holds screws and nails in wood. 1

Disadvantages (any two): friction causes wear and tear of moving parts; energy is wasted (dissipated as heat) to overcome friction in machinery. 1

Method to reduce friction (any one): polishing rough surfaces; using ball bearings (converts sliding friction to rolling friction); applying lubricants (oil or grease). 1

OR

Centripetal Force Calculation 3

Centripetal force: the force that pulls an object out of its straight-line path and into a circular path, always directed towards the centre of the circle: $F_c = mv^2/r$ 1

Given: $m = 0.6 \text{ kg}$, $r = 0.9 \text{ m}$, $v = 6 \text{ m/s}$

$$F_c = mv^2/r = (0.6)(6)^2/0.9 = (0.6)(36)/0.9 = 21.6/0.9 = 24 \text{ N} \quad 2$$

2(viii) Atmospheric Pressure and Weather 3

Atmospheric pressure varies from place to place and day to day. These variations are shown on weather maps as **isobars** — lines joining all places with the same atmospheric pressure. 1

High pressure regions (H) and **low pressure regions (L)** are marked on weather maps, and winds move from high pressure to low pressure regions, bringing changes in weather. 1

When isobars are **closely packed together**, it indicates a **large pressure difference** over a short distance, which means **stronger winds**. Widely spaced isobars indicate a smaller pressure difference and gentler winds. 1

OR

Layered Liquid Pressure (Petrol on Water) 3

Given: petrol layer: $h_1 = 2 \text{ m}$, $\rho_1 = 700 \text{ kg/m}^3$; water layer: $h_2 = 3 \text{ m}$, $\rho_2 = 1000 \text{ kg/m}^3$; $g = 10 \text{ m/s}^2$

(a) Pressure due to petrol alone: $P_1 = \rho_1 gh_1 = 700 \times 10 \times 2 = 14,000 \text{ Pa}$ 1.5

(b) Pressure due to water layer: $P_2 = \rho_2 gh_2 = 1000 \times 10 \times 3 = 30,000 \text{ Pa}$. Total pressure at the bottom = $P_1 + P_2 = 14,000 + 30,000 = 44,000 \text{ Pa}$ 1.5

2(ix) Pressure — Snowmobile 3

Pressure is the force acting perpendicularly per unit area: $P = F/A$. SI unit: **pascal (Pa)**, equal to N/m^2 . 1

Given: $F = 4500 \text{ N}$, $A = 0.25 \text{ m}^2$

$$P = F/A = 4500/0.25 = 18,000 \text{ Pa} \quad 2$$

OR

Pascal's Principle — Balloon 3

Pascal's principle: pressure applied to an enclosed fluid is transmitted equally in all directions throughout the fluid, and acts with equal force on equal areas. 1

When you blow up a **balloon**, the pressure of the air inside the balloon increases. 1

This increased pressure spreads out equally in every direction inside the balloon and pushes outward on the entire inner surface with equal force per unit area, making the balloon expand uniformly and grow bigger in all directions rather than bulging at only one point. 1

2(x) Work Done at an Angle (Toy Wagon) 3

Work done: the product of the force and the displacement moved in the direction of the force: $W = Fd$. When the force is at an angle θ to the displacement, only the component of force along the displacement does work: $W = Fd \cos\theta$. 1

Given: $F = 80 \text{ N}$, $\theta = 40^\circ$ ($\cos 40^\circ = 0.77$), $d = 6 \text{ m}$

$W = Fd \cos\theta = (80)(6)(0.77) = 369.6 \text{ J}$ 2

OR

Formation of Fossil Fuels 3

Fossil fuels are the remains of millions of years of decayed and compressed organic matter (plants and marine organisms) buried under layers of sediment. Over millions of years, heat and pressure within the Earth transform this organic matter into hydrocarbon-rich fuels. 1

Three main types: **coal** (from ancient plants), **petroleum (oil)** and **natural gas** (both mainly from marine organisms). 1

Environmental disadvantage (any one): burning fossil fuels releases **greenhouse gases (CO_2)** contributing to global warming, and produces air pollutants harmful to health. 1

2(xi) Physical vs Biological Sciences 3

Science is broadly divided into two main branches. **Physical Sciences** study non-living matter; branches include **Physics, Chemistry, Geology, Geography** (any two). 1

Biological Sciences study living organisms; branches include **Botany, Zoology, Genetics, Medicine** (any two). 1

Physics belongs to the Physical Sciences, since it is the study of non-living matter, energy, space and time. 1

OR

Physics as an Interdisciplinary Field 3

Modern science increasingly requires researchers from different fields to work together, and physics principles underpin many other disciplines besides biophysics. 1

Example 1 (Chemistry): Chemistry relies on the physics of **subatomic particles** to understand why and how chemical reactions take place. 1

Example 2 (Geology): Geology uses the physics of **waves and energy transfer** to determine the magnitude and location of earthquakes (seismology). 1

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

3(i) Derivation of $v = u + at$ & Van Kinematics 5

(a) Derivation of $v = u + at$ 2

By definition, acceleration is the rate of change of velocity: $a = (v - u)/t$, where u is the initial velocity, v is the final velocity after time t . Rearranging for v : **$v = u + at$** . (Equivalently, this can be shown graphically as the slope of a speed-time graph: slope = $a = (v - u)/t$, giving the same relation.)

(b) Numerical — Given: $u = 7 \text{ m/s}$, $v = 27 \text{ m/s}$, $s = 136 \text{ m}$ 3

(i) Acceleration: using $v^2 = u^2 + 2as \Rightarrow 27^2 - 7^2 = 2a(136) \Rightarrow 729 - 49 = 272a \Rightarrow 680 = 272a \Rightarrow a = 2.5 \text{ m/s}^2$

(ii) Time taken: using $v = u + at \Rightarrow 27 = 7 + (2.5)t \Rightarrow 20 = 2.5t \Rightarrow t = 8 \text{ s}$

OR

Three Equations of Motion & Free Fall (Coconut) 5

(a) Three equations of motion 2

- **$v = u + at$** — v : final velocity, u : initial velocity, a : acceleration, t : time
- **$s = ut + \frac{1}{2}at^2$** — s : displacement covered
- **$v^2 = u^2 + 2as$**

(b) Numerical — Given: $u = 0$, $t = 2.5 \text{ s}$, $g = 10 \text{ m/s}^2$ 3

(i) Velocity at impact: $v = u + gt = 0 + (10)(2.5) = 25 \text{ m/s}$

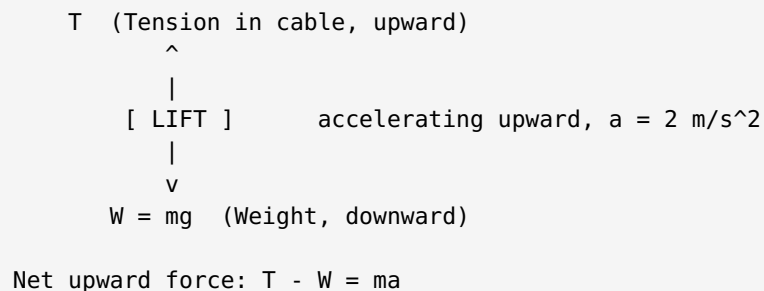
(ii) Height of tree: $h = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2}(10)(2.5)^2 = \frac{1}{2}(10)(6.25) = 31.25 \text{ m}$

3(ii) Newton's Laws, FBD of a Lift & Tension 5

(a) Newton's three laws, with examples (not involving a vehicle or firearm) 2

- **First law (inertia):** a body remains at rest or in uniform motion unless acted on by a net external force.
Example: a coin resting on a card on top of a glass stays behind when the card is flicked away quickly, falling into the glass.
- **Second law:** $F = ma$; the acceleration of a body is proportional to the net force and inversely proportional to its mass. Example: it is harder to push a loaded shopping trolley than an empty one.
- **Third law:** for every action there is an equal and opposite reaction. Example: pressing a stone with a finger — the finger feels the stone pressing back equally.

(b) Free-body diagram of the lift (accelerating upward) 1



(c) Numerical — Given: $m = 800 \text{ kg}$, $a = 2 \text{ m/s}^2$, $g = 10 \text{ m/s}^2$ 2

Applying Newton's second law along the vertical: $T - mg = ma \Rightarrow T = m(g + a) = 800(10 + 2) = 800 \times 12 = 9600 \text{ N}$

OR

Conservation of Momentum — Head-On Collision 5

(a) Law of conservation of momentum 1

In an isolated system (no external force), the **total momentum before a collision equals the total momentum after the collision**: $m_1u_1 + m_2u_2 = (m_1 + m_2)v$ (for bodies that stick together).

(b) Numerical — Given: car: $m_1 = 1000 \text{ kg}$, $u_1 = +18 \text{ m/s}$; truck: $m_2 = 1500 \text{ kg}$, $u_2 = -6 \text{ m/s}$ (opposite direction) 2

Total momentum before = $m_1u_1 + m_2u_2 = (1000)(18) + (1500)(-6) = 18,000 - 9,000 = 9,000 \text{ kg}\cdot\text{m/s}$

Common velocity: $v = 9,000/(1000 + 1500) = 9,000/2,500 = 3.6 \text{ m/s}$ (in the car's original direction)

(c) Loss in kinetic energy 2

KE before = $\frac{1}{2}(1000)(18)^2 + \frac{1}{2}(1500)(6)^2 = 162,000 + 27,000 = 189,000 \text{ J}$

KE after = $\frac{1}{2}(2500)(3.6)^2 = \frac{1}{2}(2500)(12.96) = 16,200 \text{ J}$

Loss in KE = $189,000 - 16,200 = 172,800 \text{ J (172.8 kJ)}$

3(iii) Derivation of $P = \rho gh$, Dam Wall Pressure & Force 5

(a) Derivation of $P = \rho gh$ 2

Consider a liquid column of cross-sectional area A and height h . Its volume $= Ah$, and mass $= \rho Ah$ (since $\rho = \text{mass/volume}$). Weight of this column $= mg = \rho Ahg$. This weight acts perpendicular to the base area A , so pressure at the base: $P = F/A = (\rho Ahg)/A = \rho gh$.

(b) Numerical — Given: $h = 18 \text{ m}$, $\rho = 1000 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$ 2

$$P = \rho gh = 1000 \times 10 \times 18 = \mathbf{180,000 \text{ Pa } (1.8 \times 10^5 \text{ Pa})}$$

(c) Force on dam base — Given: $A = 40 \text{ m}^2$ 1

$$F = P \times A = 180,000 \times 40 = \mathbf{7,200,000 \text{ N } (7.2 \times 10^6 \text{ N})}$$

OR

Hooke's Law & Spring Constant 5

(a) Hooke's law and spring constant 2

Hooke's law: the extension of a spring is directly proportional to the applied force, provided the elastic limit is not exceeded: $F = kx$. The **spring constant k** is the force required to produce unit extension; SI unit: **newton per metre (N/m)**.

(b) Numerical — Given: $F = 10 \text{ N}$, $x = 5 \text{ cm} = 0.05 \text{ m}$; max extension $= 12 \text{ cm} = 0.12 \text{ m}$ 3

(i) Spring constant: $k = F/x = 10/0.05 = \mathbf{200 \text{ N/m}}$

(ii) Maximum load within Hooke's law: $F_{\text{max}} = k \times x_{\text{max}} = 200 \times 0.12 = \mathbf{24 \text{ N}}$

3(iv) Conservation of Energy — Falling Rock 5

(a) Law of conservation of energy 1

Energy can **neither be created nor destroyed**; it can only be converted from one form to another. The total energy of an isolated system remains constant.

(b) Numerical — Given: $m = 6 \text{ kg}$, $h = 45 \text{ m}$, $g = 10 \text{ m/s}^2$ 3

(i) PE at the top: $E_p = mgh = (6)(10)(45) = \mathbf{2700 \text{ J}}$

(ii) KE and speed at impact (no air resistance): by conservation of energy, all PE converts to KE, so $E_k = 2700 \text{ J}$. Using $E_k = \frac{1}{2}mv^2 \Rightarrow 2700 = \frac{1}{2}(6)v^2 \Rightarrow v^2 = 900 \Rightarrow \mathbf{v = 30 \text{ m/s}}$

(c) Work done against air resistance — Given: actual impact speed $= 24 \text{ m/s}$ 1

Actual KE at impact $= \frac{1}{2}(6)(24)^2 = \frac{1}{2}(6)(576) = 1728 \text{ J}$. Work done against air resistance $= \text{PE} - \text{actual KE} = 2700 - 1728 = \mathbf{972 \text{ J}}$

OR

Fossil Fuels & Nuclear Fuel 5

(a) Formation of fossil fuels and three main types 2

Fossil fuels are formed from the remains of plants and marine organisms buried under sediment over millions of years; heat and pressure convert this organic matter into hydrocarbon-rich fuels. The three main types are **coal, petroleum (oil), and natural gas**.

(b) Nuclear fission and nuclear fuels 2

Nuclear fission is the process of splitting heavy atoms (such as uranium) into lighter nuclei, releasing a large amount of energy in the process. The two materials commonly used as nuclear fuel in power plants are **Uranium-235 (U-235)** and **Plutonium-239 (Pu-239)**.

(c) Advantage and disadvantage of nuclear fuel 1

Advantage: nuclear fuel can produce a very large amount of energy from a small quantity of fuel, and it does not emit greenhouse gases or air pollutants during operation. **Disadvantage:** it produces radioactive waste that remains hazardous for a very long time and requires careful, costly disposal.